

MATH 324 Winter 2025 - Tutorial 7

Section 003 (Thursday)

February 27, 2025

1 Maximum Likelihood Estimation

Question 1 - Pareto distribution

Let $X_1, X_2, \dots, X_n$ be a random sample with the p.d.f.

$$f(x|\theta) = \theta x^{-2}, \quad 0 < \theta \leq x < \infty.$$

- (a) What is a sufficient statistic for θ ?
- (b) Find the MLE of θ .
- (c) Find the method of moments estimator of θ .

Solution

(a)

$$f(x_1, \dots, x_n|\theta) = \prod_{i=1}^n \theta x_i^{-2} I_{(\theta, \infty)}(x_i) = \theta^n I_{(\theta, \infty)}(x_{(1)}) \prod_{i=1}^n x_i^{-2}$$

The first half (before the product) only depends on $(x_{(1)}, \theta)$ and the second half only depends on $(x_1, \dots, x_n)$. Thus, by factorization theorem, $X_{(1)}$ is sufficient.

(b)

$$L(\theta|\mathbf{x}) = \theta^n I_{(\theta, \infty)}(x_{(1)}) \prod_{i=1}^n x_i^{-2}$$

For $0 < \theta \leq x_{(1)}$, $L(\theta|\mathbf{x}) = \theta^n \prod_{i=1}^n x_i^{-2}$, which is increasing in θ .

For $\theta > x_{(1)}$, $L(\theta|\mathbf{x}) = 0$.

Putting these together, $L(\theta|\mathbf{x})$ is maximized at $\hat{\theta} = X_{(1)}$.

(c)

$$\mu_1 = \mathbb{E}[X_i] = \int_{\theta}^{\infty} x \theta x^{-2} dx = \theta \int_{\theta}^{\infty} \frac{1}{x} dx = \infty.$$

The expectation does not exist, so the M.o.M. estimator does not exist.

Question 2 - Restricted parameter Bernoulli MLE

Let $X_1, \dots, X_n$ be a random sample from a population with p.m.f.

$$\mathbb{P}(X = x|\theta) = \theta^x(1-\theta)^{1-x}, \quad x = 0 \text{ or } 1, \quad 0 \leq \theta \leq \frac{1}{2}$$

- Find the method of moments estimator and MLE of θ .
- Find the mean squared errors of each of the estimators.
- Which estimator is preferred? Justify your choice.

Solution

(a) M.O.M. $\mu_1 = E[X] = \theta \Rightarrow \hat{\theta}^{(M.O.M)} = \bar{X}$
(does not consider restriction)

MLE: $L(\theta|X) = \prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i} = \theta^y (1-\theta)^{n-y}$

Let $y = \sum_{i=1}^n x_i \rightarrow$ max at $\theta = \bar{X}$, assuming no restrictions

$L(\theta|X)$ increasing for $0 \leq \theta \leq \bar{X}$
 decreasing for $\bar{X} \leq \theta \leq 1$.



When $\bar{X} < \frac{1}{2}$, $\bar{X}$ is the MLE of θ , since $\bar{X}$ is max $L(\theta|X)$

When $\bar{X} > \frac{1}{2}$, then $L(\theta|X)$ is increasing on $(0, 1/2]$
 obtains the max at the upperbound.

$$\Rightarrow \hat{\theta} = \min \left\{ \bar{X}, \frac{1}{2} \right\}$$

(b) $MSE(\hat{\theta}^{(M.O.M)}) = \text{Var}(\hat{\theta}^{(M.O.M)}) + \left(\text{Bias}(\hat{\theta}^{(M.O.M)}) \right)^2$
 $= \frac{\theta(1-\theta)}{n} + 0 = \frac{\theta(1-\theta)}{n}$

$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$

$$= \sum_{y=0}^n (\hat{\theta} - \theta)^2 \binom{n}{y} \theta^y (1-\theta)^{n-y}$$

$$= \sum_{y=0}^{\lfloor n/2 \rfloor} \left(\frac{y}{n} - \theta \right)^2 \binom{n}{y} \theta^y (1-\theta)^{n-y} \quad \text{consider when } \bar{X} \leq \frac{1}{2}$$

$$+ \sum_{y=\lfloor n/2 \rfloor + 1}^n \left(\frac{1}{2} - \theta \right)^2 \binom{n}{y} \theta^y (1-\theta)^{n-y} \quad \text{considers when } \bar{X} > \frac{1}{2}$$

$Y \sim \text{Binomial}(n, \theta)$ and $\lfloor n/2 \rfloor = \frac{n}{2}$ if n is even
 $= \frac{n-1}{2}$ if n is odd.

(c) $MSE(\hat{\theta}^{(M.O.M)}) = E[(\bar{X} - \theta)^2] = \sum_{y=0}^n \left(\frac{y}{n} - \theta \right)^2 \binom{n}{y} \theta^y (1-\theta)^{n-y}$

$$MSE(\hat{\theta}^{(M.O.M)}) - MSE(\hat{\theta}) = \sum_{y=\lfloor n/2 \rfloor + 1}^n \left[\left(\frac{y}{n} - \theta \right)^2 - \left(\frac{1}{2} - \theta \right)^2 \right] \binom{n}{y} \theta^y (1-\theta)^{n-y} \quad \leftarrow \text{we are considering the cases when } \bar{X} > \frac{1}{2}$$

$$= \sum_{y=\lfloor n/2 \rfloor + 1}^n \underbrace{\left(\frac{y}{n} + \frac{1}{2} - 2\theta \right)}_{> 0} \underbrace{\left(\frac{y}{n} - \frac{1}{2} \right)}_{> 0} \underbrace{\binom{n}{y} \theta^y (1-\theta)^{n-y}}_{> 0 \text{ since p.m.f. of Bernoulli}} > 0 \quad \text{for } 0 < \theta \leq \frac{1}{2}$$

$$\Rightarrow MSE(\hat{\theta}) \leq MSE(\hat{\theta}^{(M.O.M)}) \rightarrow \text{MLE is preferred}$$

Question 3 - Regression MLE

Suppose that the random variables $Y_1, \dots, Y_n$ satisfy

$$Y_i = \beta x_i + \varepsilon_i, \quad i = 1, \dots, n$$

where $x_1, \dots, x_n$ are fixed constants $\varepsilon_1, \dots, \varepsilon_n$ are i.i.d. $N(0, \sigma^2)$, σ^2 unknown.

- Find a two dimensional sufficient statistic for (β, σ^2) .
- Find the MLE of β , and show that it is an unbiased estimator of β .
- Find the distribution of the MLE of β .

Solution

(a) $Y_i \sim N(\beta x_i, \sigma^2)$.

$$\begin{aligned}
 L(\beta, \sigma^2 | \mathbf{y}) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ \frac{-1}{2\sigma^2} (y_i - \beta x_i)^2 \right\} \\
 &= (2\pi\sigma^2)^{n/2} \exp \left\{ \frac{-1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta x_i)^2 \right\} \\
 &= (2\pi\sigma^2)^{n/2} \exp \left\{ \frac{-1}{2\sigma^2} \sum_{i=1}^n (y_i^2 - 2\beta x_i y_i + \beta^2 x_i^2) \right\} \\
 &= (2\pi\sigma^2)^{n/2} \exp \left\{ \frac{-1}{2\sigma^2} \sum_{i=1}^n y_i^2 + \frac{\beta}{\sigma^2} \sum_{i=1}^n x_i y_i \right\} \exp \left\{ \frac{-\beta^2 \sum_{i=1}^n x_i^2}{2\sigma^2} \right\}
 \end{aligned}$$

By factorization theorem, $(\sum_{i=1}^n Y_i^2, \sum_{i=1}^n x_i Y_i)$ is a sufficient statistic.

(b)

$$\log L(\beta, \sigma^2) = \frac{-n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n y_i^2 + \frac{\beta}{\sigma^2} \sum_{i=1}^n x_i y_i - \frac{\beta^2}{2\sigma^2} \sum_{i=1}^n x_i^2$$

Fix σ^2 .

$$\frac{\partial L}{\partial \beta} = \frac{1}{\sigma^2} \sum_{i=1}^n x_i y_i - \frac{\beta}{\sigma^2} \sum_{i=1}^n x_i^2$$

Set $\frac{\partial L}{\partial \beta} = 0$.

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

Note that $\frac{\partial^2 L}{\partial \beta^2} = \frac{-1}{\sigma^2} \sum_{i=1}^n x_i^2 < 0$, so $\hat{\beta}$ is a max.

$$\mathbb{E}[\hat{\beta}] = \frac{1}{\sum_{i=1}^n x_i^2} \sum_{i=1}^n x_i \mathbb{E}[Y_i] = \frac{1}{\sum_{i=1}^n x_i^2} \sum_{i=1}^n x_i \beta x_i = \beta$$

Therefore, $\hat{\beta}$ is unbiased.

(c) $\hat{\beta} = \sum_{i=1}^n a_i Y_i$ where $a_i = \frac{x_i}{\sum_{j=1}^n x_j^2}$ and Y_i . $\hat{\beta}$ is the linear combination of Normal random variables and thus is also Normally distributed.

$$\text{Var}(\hat{\beta}) = \sum_{i=1}^n a_i^2 \text{Var}(Y_i) = \sum_{i=1}^n a_i^2 \sigma^2 = \frac{\sigma^2}{\sum_{j=1}^n x_j^2}$$

Therefore, $\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{\sum_{j=1}^n x_j^2}\right)$

2 MLE Asymptotics

2.1 Score function

Definition: The *score function* is

$$\frac{\partial}{\partial \theta} \log f(X; \theta)$$

which we've seen for finding the MLE. By chain rule and certain regularity conditions (to swap the derivative and the integral), it can be shown that

$$\mathbb{E} \left[\frac{\partial}{\partial \theta} \log f(X; \theta) \middle| \theta \right] = 0.$$

2.2 Fisher information

Definition: The *Fisher information*, $\mathcal{I}(\theta)$, is the variance of the score function. Under regularity conditions (most of the distributions we'll see in class),

$$\mathcal{I}(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \middle| \theta \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \middle| \theta \right]$$

The Fisher information measures the amount of 'information' X carries about θ . For instance,

- If the likelihood sharply peaks with respect to changes in θ , this suggests X provides a lot of information about θ ;
- A flatter likelihood would need a larger sample to properly estimate the true value of θ .

The Fisher information seeks to capture this 'variance' with respect to θ .

2.3 Asymptotic normality of MLE

Theorem Suppose $\hat{\theta}$ is a consistent estimator for θ and t is a one-to-one function (by invariance property of MLE, $t(\hat{\theta}) \xrightarrow{p} t(\theta)$). Then,

$$Z = \frac{t(\hat{\theta}) - t(\theta)}{\sqrt{\left[\frac{\partial t(\theta)}{\partial \theta} \right]^2 / n \mathcal{I}(\theta)}}$$

has a standard Normal distribution. We can view this as:

$$\sqrt{n}(t(\hat{\theta}) - t(\theta)) \sim N \left(0, \left[\frac{\partial t(\theta)}{\partial \theta} \right]^2 \mathcal{I}^{-1}(\theta) \right)$$

or, if $t(\theta) = \theta$,

$$\hat{\theta} \sim N \left(\theta, \frac{\mathcal{I}^{-1}(\theta)}{n} \right).$$

This means Z is a pivotal quantity that can be used for asymptotic confidence intervals of $t(\theta)$.

Question 4 - Confidence interval from Poisson

(Exercise 9.101) Let $Y_1, Y_2, \dots, Y_n$ denote a random sample of size n from a Poisson distribution with mean λ . Find a $100(1 - \alpha)\%$ confidence interval for $t(\lambda) = e^{-\lambda} = P(Y = 0)$.

Solution

We have that the MLE for λ is $\bar{Y}$. Thus by the invariance property, the MLE for $t(\lambda)$ is $\exp(-\bar{Y})$.

To find the Fisher information, we look for the second derivative of $\log p(Y; \lambda)$.

$$\begin{aligned} p(y; \lambda) &= e^{-\lambda} \frac{\lambda^y}{y!} \\ \log p(y; \lambda) &= -\lambda + y \log \lambda - \log(y!) \\ \frac{d}{d\lambda} \log p(y; \lambda) &= -1 + \frac{y}{\lambda} \\ \frac{d^2}{d\lambda^2} \log p(y; \lambda) &= \frac{-y}{\lambda^2} \end{aligned}$$

So the Fisher information is

$$-\mathbb{E} \left[\frac{d^2}{d\lambda^2} \log p(Y; \lambda) \middle| \lambda \right] = -\mathbb{E} \left[\frac{-Y}{\lambda^2} \right] = \frac{1}{\lambda^2} \mathbb{E}[Y] = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

As well,

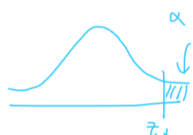
$$\frac{d}{d\lambda} t(\lambda) = -e^{-\lambda}$$

our approximate $100(1 - \alpha)\%$ confidence interval is

$$\begin{aligned} t(\hat{\lambda}) \pm z_{\alpha/2} \sqrt{\left[\frac{\partial t(\theta)}{\partial \theta} \right]^2 \bigg/ n \mathcal{I}(\theta)} &= \exp(-\bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{e^{-2\lambda}}{n/\lambda} \bigg|_{\lambda=\bar{Y}}} \\ &= \exp(-\bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{\bar{Y} e^{-2\bar{Y}}}{n}} \end{aligned}$$

3 Hypothesis Testing

$\leftarrow$ usually try to reject/disprove H_0 (e.g. $\mu=0$, $\mu_1=\mu_2$)
 $\rightarrow$ unknown $\rightarrow \theta = \theta_0$ known, fixed, e.g. $\theta_0=0$
 1) H_0 : null
 2) H_a (H_1): alternative $\theta > \theta_0$ (or $\theta < \theta_0$ or $\theta \neq \theta_0$)
 3) Test statistic: $\hat{\theta}$
 4) Rejection region: Reject H_0 if $\{\hat{\theta} > k\}$ for some k
 Instead of using $\hat{\theta}$, we usually use $Z = \frac{\hat{\theta} - \theta_0}{\sigma_{\hat{\theta}}}$ (so that we can use $N(0,1)$ tables)



Rejection region: $H_a: \theta > \theta_0 \Rightarrow RR = \{Z > z_\alpha\}$ (one-sided tests)
 $H_a: \theta < \theta_0 \Rightarrow RR = \{Z < -z_\alpha\}$
 $H_a: \theta \neq \theta_0 \Rightarrow RR = \{|Z| > z_{\alpha/2}\}$ (two-sided (two-tailed) test)

RESULTS	REALITY	
	H_0 true	H_0 false
accept H_0	✓	Type II error β
reject H_0	Type I error α	✓

✓ - correct decision (ie. conclusion aligns with truth, no error)

We control our error by having Type I error be max α
 i.e. $P(\text{Type I}) = \alpha$

$\text{Power} = 1 - \beta = 1 - P(\text{Type II})$
 $\alpha \downarrow, \beta \uparrow$

Question 5 - Hypothesis testing

(Exercise 10.4) Suppose that we wish to test the null hypothesis H_0 that the proportion p of ledger sheets with errors is equal to 0.05 versus the alternative H_a , that the proportion is larger than 0.05, by using the following scheme. Two ledger sheets are selected at random. If both are error free, we reject H_0 . If one or more contains an error, we look at a third sheet. If the third sheet is error free, we reject H_0 . In all other cases, we accept H_0 .

- In terms of this problem, what is a type I error?
- What is the value of α associated with this test?
- In terms of this problem, what is a type II error?
- Calculate $\beta = \mathbb{P}(\text{type II error})$ as a function of p .

Solution

This question was not covered this week! Hypothesis testing will be covered next tutorial.